

PG 1st Semester Examination, 2023 (R&B)

Subject : MCA

Paper — 1.1.2

Full Marks : 80

Time : 3 hours

Answer all questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

1. (a) Define Number system. Briefly explain about types of number system with example and their representation style. 20

Or

- (a) Design 94-nit up/down counter using a negative edge triggered T-flip-flop and the clock is applied in an asynchronous manner. 10

(Turn Over)

(b) Design a BCD to 7-segment and display a decoder circuit using logic gates. 10

2. (a) What is the use fo Adder ? Design a half-adder and full-adder ? 20

Or

(b) Define combinational logic circuit. Explain the components of combinational RTL. 20

3. (a) Define cache memory. Briefly explain about types of mapping functions of cache memory with example. 20

Or

(b) Define instruction set and its type with example. What is opcode and operand in an instruction ? 20

4. (a) Define pipeline and its advantages. Explain arithmetic pipeline and instruction pipeline with example. 20

(3)

Or

- (b) Write down the characteristics of multi-processor systems. Explain interconnection networks of multi processor systems in detailed manner.

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